



<https://doi.org/10.1057/s41599-026-07310-z>

OPEN

# Retraction Note: The effect of ChatGPT on students' learning performance, learning perception, and higher-order thinking: insights from a meta-analysis

Jin Wang & Wenxiang Fan

Retraction Note to: *Humanities and Social Sciences Communications* <https://doi.org/10.1057/s41599-025-04787-y>, published online 06 May 2025

The Editor has decided to retract this paper owing to concerns regarding discrepancies in the meta-analysis. These issues ultimately undermine the confidence the Editor can place in the validity of the analysis and resulting conclusions. The authors have not responded to correspondence regarding this retraction.

Published online: 22 April 2026



**Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026